

Notes of fusion

Xin-Peng Li

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1 Finite-temperature; 0-density

1.1 $D + T \rightarrow \alpha + n$

$D + T \rightarrow \dots$, mass m_D, m_T , reduced mass

$$\mu = \frac{m_D m_T}{m_D + m_T}. \quad (1)$$

the reaction rate per unit volume for the fusion of two nuclei

$$n_{DT} = n_D n_T \langle \sigma v \rangle, \quad (2)$$

¹ where n_D and n_2 are the densities of the two nuclei, $\langle \sigma v \rangle$ is the reaction probability.

$$\begin{aligned} \langle \sigma v \rangle &= \int_0^{+\infty} n(v) v \sigma(v) dv \\ &= \mu^{-1} \int_0^{+\infty} n(E) \sigma(E) dE \end{aligned} \quad (3)$$

where $v(E) = \sqrt{2E/\mu}$
Maxwell-Boltzmann distribution:

$$n(E) = \frac{2E\sqrt{2\mu}}{\sqrt{\pi kT} \mu kT} e^{-E/kT} \quad (4)$$

So

$$\langle \sigma v \rangle = \frac{2\sqrt{2}}{\sqrt{\pi kT} \mu kT} \int_0^{+\infty} E e^{-E/kT} \sigma(E) dE \quad (5)$$

Using astrophysical S factor

$$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta(E)}, \quad (6)$$

where $\eta(E) = \frac{Z_D Z_T e^2}{\hbar} \sqrt{\frac{\mu}{2E}}$
So

$$\langle \sigma v \rangle = \frac{2\sqrt{2}}{\sqrt{\pi kT} \mu kT} \int_0^{+\infty} S(E) e^{-E/kT - 2\pi\eta(E)} dE \quad (7)$$

We could calculate $\langle \sigma v \rangle(T)$, A reference figure is shown HERE.(Another reference figure here)
REFERENCE ARTICLE.

Table 1: Input parameters for calculating $\langle \sigma v \rangle(T)$

Sign	Num
Z_D, Z_T	1
AME	0.5110034D0 MeV/c ²
amp	938.2720813d0 MeV/c ²
amn	939.5654133d0 MeV/c ²
$AMC12$	931.494D0 MeV/c ²
m_D	3670.481D0 * AME
m_T	5496.918D0 * AME
$AMALP$	4.001506D0 * AMC12
$\frac{e^2}{\hbar}$	1/137 c
k	1
$S(E)$	$\frac{26 - 0.361E + 248E^2}{1 + (\frac{E - 0.0479}{0.0392})^2} \text{ MeV} \cdot b$

$S(E)$ using equation from HERE.

¹If the two nuclei are identical, e.g. $D + D$, then $n_{DD} = \frac{1}{2} n_D^2 \langle \sigma v \rangle$

Table 2: Output parameters

Sign	Unit
T	$1MeV = 10^3 KeV$
$\langle\sigma v\rangle$	$1b \cdot c \approx 1.807 * 10^{10} cm^3/(Mol \cdot s)$

when integrate to $E = 300KeV$, the result is shown below.

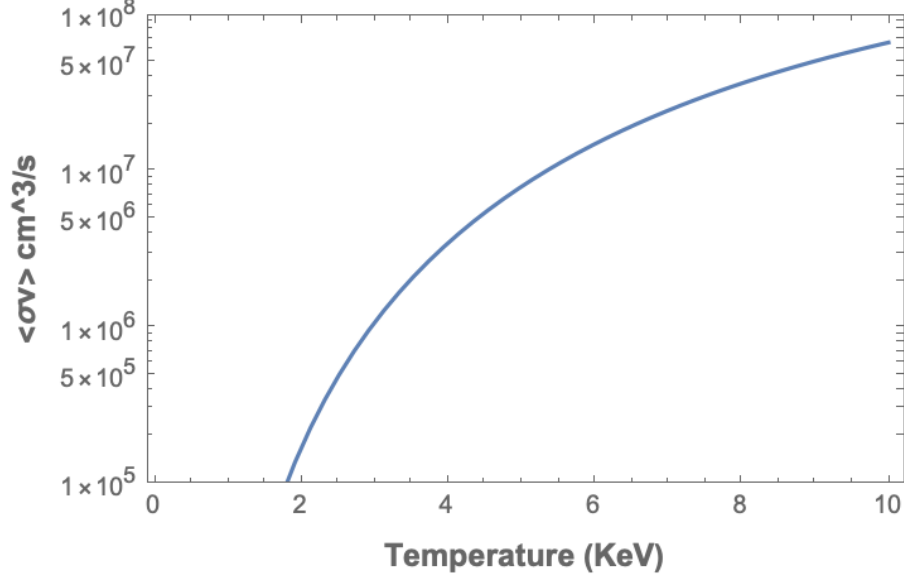


Figure 1: $\langle\sigma v\rangle(T)$ for $D + T \rightarrow \alpha + n$

The result is very close to the reference figure.3 HERE.

- The reason that the range for T is from $0.1KeV$ to $10KeV$ is that keep $\frac{E_{uplim}}{kT} \lesssim 30$
- More range needs increase E_{uplim}

1.2 $D + T + \mu^- \rightarrow \alpha + n + \mu^-$

1.2.1 Major process

First

$$\begin{aligned} \mu &\xrightarrow{10^{-12}s} d\mu \xrightarrow{10^{-9}s} t\mu \\ &\rightarrow t\mu \end{aligned} \quad (8)$$

Second

$$t\mu + dd \xrightarrow{10^{-9}s} dt\mu + d \quad (9)$$

Third

$$dt\mu \xrightarrow{10^{-12}s} \alpha + n + \mu + 17.6MeV \quad (10)$$

Fourth, over 99.6% of μ repeats cycle, 0.4% of sticking loss $\alpha\mu$.

μ lifetime $\tau_\mu = 2.2 \times 10^{-6}s$ is very long compared to above processes, so one μ could catalyze many times fusion, more than $\frac{10^{-6}}{10^{-8}} = 100$ times.

1.2.2 Reaction rate

The molecular-formation rate(per $t\mu$) is

$$\lambda_{dt\mu} = \pi \rho_d \Sigma_{K_i K_f} P_{K_i}(T) \int_0^\infty dE |\langle \Psi_f | H_{int} | \Psi_i \rangle|^2 f_{cm}(E; T) \delta(E - E_{K_i K_f}^{res}) \quad (11)$$

where $\Psi_i = |[dx]_{K_i 0}, (t\mu)_{1s}^F\rangle$ is the initial state wave function and $\Psi_f = |[(dt\mu)_{11}^{S+} x2e]_{K_f 2}^*\rangle$ is the final state wave function. K_i and K_f is the molecular rotational angular momentum. $f_{cm}(E; T)$ obey Maxwell-Boltzmann distribution

$$f_{cm}(E; T) = \frac{2}{\sqrt{\pi}} \frac{1}{(k_B T)^{\frac{3}{2}}} \sqrt{E} e^{-E/k_B T} \quad (12)$$

And $P_{K_i}(T)$ is the occupation probability of the initial rotational state K_i at temperature T . At a form

$$P_{K_i}(T) = \frac{(2K_i + 1)\xi(K_i)e^{-E_{K_i}/k_B T}}{\Sigma_{K_i}(2K_i + 1)\xi(K_i)e^{-E_{K_i}/k_B T}} \quad (13)$$

where because of total symmetry under dd exchange.

$$\xi(K_i) = \begin{cases} \frac{1}{3}, & \text{for } K_i = \text{odd} \\ \frac{2}{3}, & \text{for } K_i = \text{even} \end{cases} \quad (14)$$

And the formation probability is

$$P_{dt\mu}(E) = \pi \Sigma_{K_i K_f} P_{K_i}(T) \int_0^\infty dE |\langle \Psi_f | H_{int} | \Psi_i \rangle|^2 f_{cm}(E; T) \delta(E - E_{K_i K_f}^{res}) \quad (15)$$

We have used the resonance approximation to get the above equation. If we consider the finite width of the resonance, we could replace the delta function with a Lorentzian

$$\delta(E - E_{K_i K_f}^{res}) \rightarrow \frac{\Gamma_{K_i K_f}/2\pi}{(E - E_{K_i K_f}^{res})^2 + \Gamma_{K_i K_f}^2/4} \quad (16)$$

If considering the below-threshold resonances, we could use Lorentzian and get

$$P_{dt\mu}(E) = \Sigma_{K_i K_f} P_{K_i}(T) \int_0^\infty dE |\langle \Psi_f | H_{int} | \Psi_i \rangle|^2 f_{cm}(E; T) \frac{\Gamma_{K_i K_f}/2}{(E - E_{K_i K_f}^{res})^2 + \Gamma_{K_i K_f}^2/4} \quad (17)$$

Note that $E_{K_i K_f}^{res}$ is the resonance energy(differences between the final and initial state internal energies), and $\Gamma_{K_i K_f}$ is the width of the resonance.

We have

$$\begin{aligned} E_{K_i K_f}^{res} &= E_{(dt\mu)_{11}} - E_{t\mu+d_2} \\ &\quad + \delta E_{dt\mu} - \delta E_{t\mu} \\ &\quad + E_{K_f} - E_{K_i} \end{aligned} \quad (18)$$

- $E_{(dt\mu)_{11}} = -660.3$ MeV(Kamimura 1988) is the binding energy of the $dt\mu$ molecular state $(dt\mu)_{11}$ compared with the $t\mu + d_2$ threshold; hence $E_{t\mu+d_2} = 0$ MeV.
- The hyperfine energy shifts of the $t\mu$ is denoted by $\delta E_{t\mu}$. We have $(t\mu)_{1s}^{F=0} = +177.9$ MeV; $(t\mu)_{1s}^{F=1} = -59.3$ MeV.(Leon, 1984)
- The hyperfine energy shifts of the $dt\mu$ is denoted by $\delta E_{dt\mu}$. We have $(dt\mu)_{1s}^{F=0} = -596 \pm 2 \pm 20$ MeV; $(dt\mu)_{1s}^{F=1} = -640 \pm 2 \pm 10$ MeV.(JOHANN, 1992)
- E_{K_f} and E_{K_i} denote the energies of the final and d_2 rovibrational states relative to the $d+d$ dissociation energy, respectively.($t\mu$ is atomic state, so no rotational energy.)
The table is shown in below.

Table 3: Rotational and vibrational energy of the $(dt\mu)_{11}d2e_{K_f2}$

K_f	value (MeV)
0	613.1
1	617.9
2	627.7
3	642.2
4	661.5
5	685.6
6	714.3
7	747.5
8	785.2
9	827.2
10	873.4

Table 4: Rotational and vibrational energy of the $dd2e$

K_i	value (MeV)
0	-31.5
1	-26.3
2	-16.3
3	-1.3
4	18.7
5	43.5
6	73
7	107.3
8	146.1
9	189.5
10	237.1

Temperature and density dependence is shown HERE in Eq.9.

1.2.3 Information in experiments

1. Temperature Dependence of Formation Rate

- For high-tritium (20-30%) D-T mixtures, the $dt\mu$ molecular formation rate increases with temperature.
- The formation rate does not peak at 540 K, but continues to rise at the highest measured temperature of 800 K, with theoretical predictions suggesting a maximum around 1,200 K.
- For low-tritium (10%) mixtures, almost no temperature dependence is observed, because the μ -transfer reaction $d\mu + t \rightarrow t\mu + d$ dominates the cycle, which is nearly temperature-independent.

2. Dependence on Host Molecules

- The $dt\mu$ molecular formation rate depends on the type of host molecule. For example, the reaction $t\mu + D_2 \rightarrow (dt\mu)D_2e$ proceeds faster than $t\mu + DT \rightarrow (dt\mu)TDe$
 - This host dependence arises from differences in the available energy levels of the two compound molecules, consistent with predictions of the resonance model.
3. Density Dependence of Formation Rate
- Certain reactions (e.g., three-body collisions like $t\mu + D_2 + T_2 \rightarrow [(dt\mu)D_2e]^* + T'_2 + \Delta E$) increase faster than linearly with density, indicating that three-body collisions play an important role in resonant $dt\mu$ formation.
 - Other reactions, such as those involving low-tritium mixtures, show a linear dependence on density.
4. Cold Fusion Characteristics
- μ -catalyzed fusion relies on intact molecules at high temperatures; if hydrogen molecules decompose, the formation rate drops sharply.
 - It cannot be used with plasmas and is therefore unsuitable for thermonuclear weapons.
 - Consequently, the term “cold fusion” is appropriate for this process.

The experiment article is [HERE](#).

2 IFMCF

First article listed HERE.